Flame Guard Project Proposal

🧯 Save the Professor’s Life When Fires · W19 (Endicott) Network Upgrade & Networked Fire Alarm
Computer Networks (PBL)
Prof. Edward Youngil Kim
Presented @November 11, 2026 11 November 2026
15 min incl. Q&A

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1. Cover Page

Field	Content
Course	Computer Networks (PBL) · Prof. Edward Youngil Kim
Project title	Save the Professor’s Life When Fires - W19 (Endicott) Network Upgrade & Networked Fire Alarm
Team name	Flame Guard
Presentation date	@November 11, 2026 11 November 2026
Cooperation history	🔥 Flame Guard Project Proposal

Members & Division of Roles

Member	Technical role	Administrative role
Pushkal	System Integration & Testing (combining all parts, 3D model of W19, fire-scenario simulation)	Leader & Report Writer
Emon	Wireless N/W (Wi-Fi coverage and capacity, WLC/AP design, sensor-to-gateway communication)	Schedule & Task Tracker (deadlines, progress follow-up)
Sadat	Wired N/W (core/distribution/access backbone, VLANs, cabling, alarm wiring)	Budget & Equipment Manager (parts list, cost estimate)
Islam	IoT Sensor & Hardware (smoke/heat detectors, alarms, sirens, microcontroller); W19 2D survey maps	Co-Leader & Meeting Note
Hassan	Alert & Monitoring Software (IoT registration server, alert rules, notifications, dashboard)	Presentation Maker

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2. Project Summary

Task

Recently, the wired and wireless network of the Endicott building (W19) has slowed down severely. We will check the current network configuration, design a new network with Cisco's latest products, and install fire detection, alarm and siren devices in every professor's office to save professors' lives when a fire occurs.

Core issues

1. **Performance:** the wired and wireless network in W19 is slow, and we do not yet know whether the bottleneck is the access layer, the Wi-Fi, or the uplink.
2. **Safety:** W19 has smoke detectors in its corridors, but they are not connected to the IP network. Nobody is notified remotely, and there is no per-office alarm or siren for the 13 professors' offices (221–232 and the Dean's office 220).
3. **Validation:** even with a good design, we need a way to prove that the alarm still reaches every office while a fire is damaging part of the building. Packet Tracer alone cannot show this (see 3.6).

Hypothesis & resolution direction

We think the problem is caused by...	So we will...
Outdated or overloaded access layer: old access switches (100 Mbps ports, or 1 Gbps uplinks shared by too many ports) and the ceiling "hub" units on 2F are congested when many users connect at once.	Replace them with current Cisco PoE+ access switches with Gigabit ports and multigigabit/10G uplinks to a collapsed core.
Wi-Fi capacity and coverage: only 5 APs identified on about 2,100 m^2 of 2F (≈ 1 AP per 420 m^2), with a single AP covering the long north/east corridor.	Add and upgrade to Wi-Fi 6 APs where speed tests show weak coverage, managed centrally by a wireless LAN controller.
Fire-safety gap: detection exists only in corridors and is not networked; offices have no alarm or siren and no remote notification.	Put a smoke detector, alarm and siren in each professor's office on a dedicated fire-safety IoT VLAN with

We think the problem is caused by...	So we will...
	QoS priority, linked to an alert server that notifies everyone.
Validation gap: Packet Tracer shows that packets flow, but not whether the network survives a real fire (rooms burning, cables and devices failing).	Use our 3D model of W19 to simulate fire scenarios and check that every office still receives the alarm; feed the results back into the Packet Tracer design.

Resolution direction (four steps)

1. **Survey** – map the current network and fire devices of W19 (done for 1F/2F; network closet still to visit).
2. **Design** – build the upgraded hierarchical network with VLANs, Wi-Fi 6 and the IoT fire chain in Cisco Packet Tracer.
3. **Validate** – run fire scenarios on the 3D model of W19 and measure whether every office is alerted.
4. **Cost** – produce an itemised bill of materials and total upgrade cost.

Technical vs. learning objectives

Technical objectives (what we will build)	Learning objectives (what we will learn)
Redesign and upgrade the W19 wired & wireless network with current Cisco products in Packet Tracer	Hands-on switch, router, WLC and AP configuration (VLANs, trunking, inter-VLAN routing, DHCP)
A smoke detector, alarm and siren in each of the 13 professors' offices, on a dedicated IoT VLAN	Hierarchical campus design (core / distribution / access) from the Cisco Campus LAN & WLAN Design Guide
A fire-scenario simulation on the 3D model of W19 that tests the chosen network	Integrating IoT devices into an enterprise network, and designing for resilience and failure
An itemised cost estimate for the upgrade	Real-world survey, documentation and teamwork (Notion cooperation history)

3. Background & Problem Description

3.1 Background

W19, the Endicott Building (Start-up Hall), has three floors built around a central courtyard (about 54 × 54 m footprint, courtyard about 24 × 33 m).

- **1F** – double-height Start-up Hall, lecture and project rooms 101–111, 112 Start-up Center, an elevator and WC block.
- **2F** – the professors' offices 221–232 on the south corridor, the Dean of ECIS (220), the AI Research Institute (201, 24 workstations, five labs), offices 202–217, the TA / International Administrative Center (19 PCs) and a conference room.
- **3F** – Roof garden (not surveyed; out of our main scope).

The building is used by students, faculty, start-up teams and administrative staff throughout the day and evening. Professors often work alone in closed offices late in the day, when corridors are empty. A corridor smoke detector that rings a local bell does not guarantee that a professor behind a closed door is warned in time, and nobody outside the building is told.

3.2 Current network (as-is)

Item	What we found	Source / how verified
Floors & room layout	3 floors; 1F rooms 101–112; 2F rooms 201–232 (no room 219); 3F playground/roof garden	Site survey, 1F directory sign, 2F emergency exit map
Professors' offices	12 offices (221–232) + Dean's office 220 = 13 offices to protect	Site survey
Internet uplink / ISP	Unknown – network closet not accessed yet	To verify with IT office
Core / distribution	Unknown	To verify with IT office
Access layer	Two ceiling units on 2F labelled 02F_HUB_A / WSU_1902_A and 02F_HUB_B / WSU_1902_B (Umbrix brand) feed nearby APs; model and whether they are hubs or switches still unknown	Photos

Item	What we found	Source / how verified
Wireless APs	5 Cisco APs identified on 2F (WSU_W19_2F_33D8, _4008, _2C00, _4280, _2860); APs seen on 1F but not yet located; models unknown	Photos + 3D model
Cabling	Category not yet confirmed	To check cable print in ceiling
IP addressing / VLANs	Unknown; assumed shared campus network	To verify
End devices	≈ 103 PCs on 2F (201: 24, TA office: 19, 215–217: 18, professors' offices: 16) and 24 on 1F, plus AV (projectors, lecture capture, speakers)	Site survey + 3D model
Security devices	5 dome CCTV cameras and 16 motion sensors on 2F; 7 motion sensors on 1F	Site survey + 3D model
Fire-safety devices	47 smoke detectors on 2F and 7 on 1F, mostly in corridors; hydrant cabinets with fire bell, call point and lamp; extinguishers; sprinklers. Conventional (not IP-networked). Fire panel location and in-office detectors still to verify	Site survey + 3D model

3.3 W19 survey maps

Islam drew the 2D survey maps below from our site visits. Each room is marked with colour-coded counts of the devices we found (legend F / W / S / C / M: fire extinguisher, Wi-Fi AP, smoke detector, computer, motion sensor). On 2F, the red arrows show the evacuation routes to the stairwells from the emergency exit map.

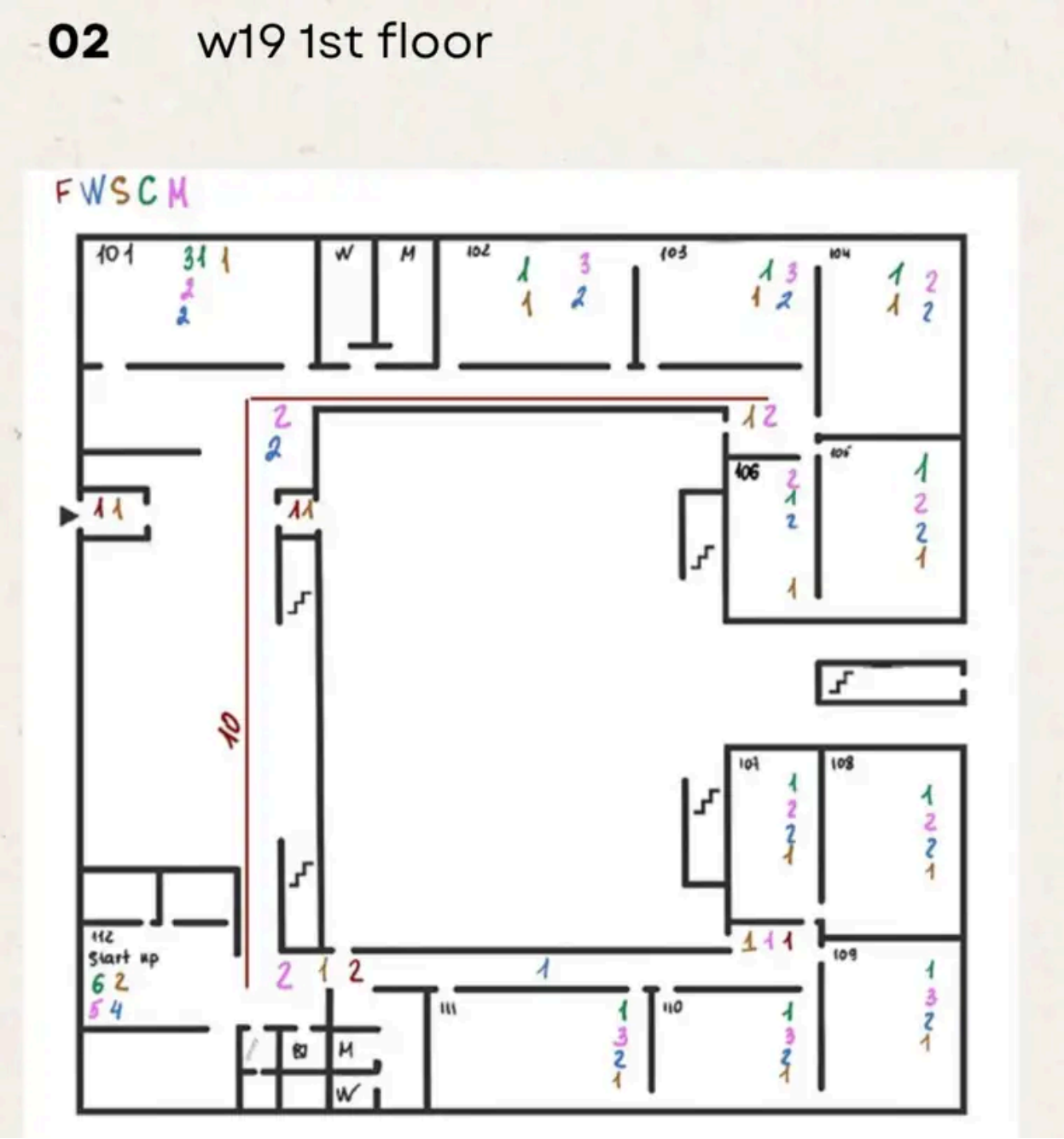


Figure 1 – W19 1F survey map (2D, by Islam): rooms 101–112 with colour-coded device counts per room.

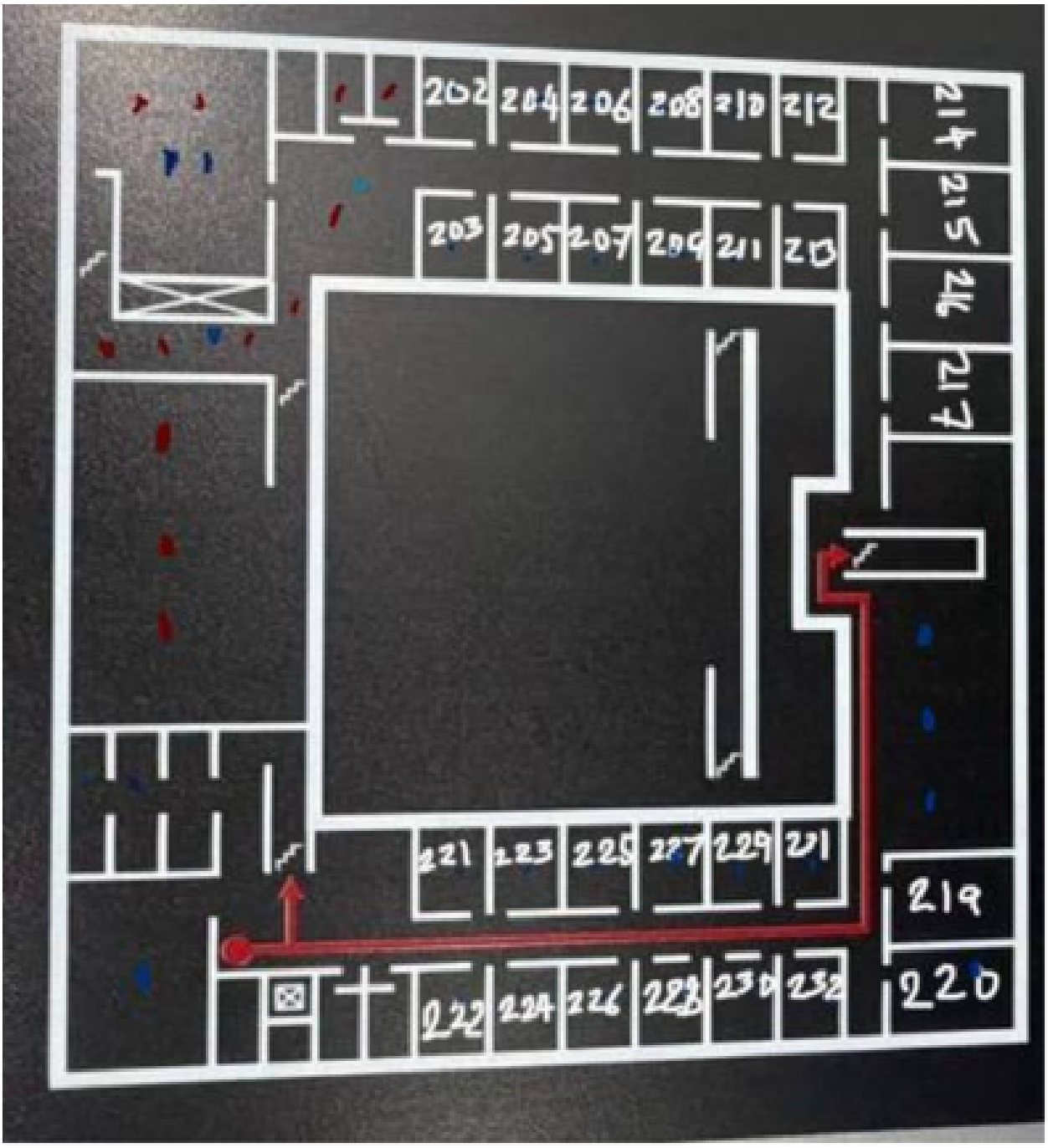


Figure 2 – W19 2F survey map (2D, by Islam): offices 201–232 with devices marked as coloured dots and evacuation routes (red arrows) to the stairwells.

3.4 Our analysis so far

From the survey, the maps and the device counts in our 3D model (373 items placed), five findings shape the design:

1. **Wi-Fi is thin for the load.** 2F has about 2,100 m² of floor (footprint minus courtyard) and only 5 identified APs, roughly 1 AP per 420 m². Demand is concentrated in a few rooms: 201 (24 workstations), the TA office (19 PCs), 215–217 (18 PCs) and the 1F lecture rooms, plus every occupant's phone and laptop. This supports hypothesis H2; speed tests will confirm it.
2. **Single point of failure on the main corridor.** AP 4280 is the only AP along the long north/east corridor, and it sits next to Hub B. One failed device, or a fire near that ceiling, removes wireless along the main evacuation route.
3. **Fire detection is present but isolated.** 47 smoke detectors on 2F, mostly in corridors (4 alone in the corridor before the west stairs), but none we have seen is networked. There is no remote notification and no confirmation that each office was alerted.
4. **The professors' offices depend on a fast alarm.** The 12 offices line the south corridor, with exits only at the corridor ends (south-west stair and east side). An alarm that reaches them seconds earlier matters most here.
5. **The biggest unknown is the core.** We have not yet seen the network closet, uplink or switch models. This is the top item in Section 4.

3.5 The problem from a network perspective

Layer	Symptom / risk in W19	Likely cause (hypothesis)
Physical (L1)	Slow wired access; unknown cable category	Old cabling or 100 Mbps ports; shared ceiling "hub" units
Data link (L2)	Congestion at peak times; IoT, CCTV and user traffic mixed	Hubs/unmanaged switches; flat network without VLAN segmentation
Network (L3)	Unknown uplink capacity; no traffic priority	Oversubscribed uplink to campus; no QoS for safety traffic
Wireless	Weak signal and slow speeds in dense rooms and the north/east corridor	Too few APs (≈ 1 per 420 m²), older Wi-Fi standard, channel interference
Application / IoT	No remote fire alerts; no per-office alarms	Fire devices not integrated with the IP network

Main bottleneck (current hypothesis): the access layer and Wi-Fi capacity on 2F, to be confirmed by speed tests and the closet visit.

3.6 Limitations of existing approaches: why Packet Tracer alone is not enough

Cisco Packet Tracer is our main design tool and the required deliverable. Looking at its current limitations, we realised that it may not be sufficient for us to simulate the event of a fire. Packet Tracer shows a logical topology and can trigger an IoT smoke detector, but it cannot show what a fire physically does to the network: which rooms burn, which ceiling devices and cable runs fail, in what order, and whether the alarm still has a path to every office.

Question we need to answer	Packet Tracer	3D model of W19
Do VLANs, routing, DHCP and the WLC work?	✅ Yes – this is its strength	❌ No (not a protocol simulator)
Does a smoke detector trigger the siren?	✅ Yes, logically	✅ Yes, in the real room location
Where exactly are devices and cables in the building?	⚠️ Only a rough physical view	✅ Every device at its real position and height
What fails when a fire spreads from room X?	❌ Not modelled	✅ Fire spread over time, devices and links in the fire zone fail
Does every office still get the alarm during the fire?	❌ Not testable	✅ Path check from each detector to every siren
Where are the single points of failure?	⚠️ Only by manual inspection	✅ Found automatically per scenario

So **Pushkal has built a 3D model of W19**, and we plan to use it to simulate events that may occur during a fire and to check whether the network configuration we chose actually works.

What the 3D model already does (source on GitHub):

- Traced floor plans of 1F, 2F and 3F (54 × 54 m footprint, rooms, walls, stairs and voids) built in three.js.
- 373 items placed from the survey: PCs, APs, hubs, smoke detectors, motion sensors, CCTV, extinguishers, hydrant cabinets, projectors, speakers and furniture.

- Each device stores type, floor, position, height, status (existing / proposed / to remove), notes, photo reference and cable links (Ethernet, fibre, wireless, fire-alarm wiring).
- Overview mode (rotate, zoom, filter by category) and walk mode (first/third person, stairs between floors, minimap).
- Export/import of the whole layout as JSON, which we will also use for device counts and the cost estimate.

How we will use it (fire-scenario simulation):

1. Place the proposed network on the model: switches, APs, WLC, closets, cable routes and the fire devices in all 13 offices.
2. Add a fire-scenario mode: choose a start room, let the fire spread room by room over time, and mark devices and links inside the fire zone as failed.
3. After every step, check the alarm path: can each detector still reach the alert server, and can the server still reach the siren in every office?
4. Measure the results: time to alert, percentage of offices alerted, and every single point of failure found.
5. Feed the results back into Packet Tracer (for example, a redundant uplink or moving a closet) and re-run.

Packet Tracer proves the network is **configured** correctly; the 3D model shows whether it **survives** a fire. Together they validate the design.

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4. Things to Investigate

4.1 Items to verify

#	Item to verify	Why it matters	Method	Owner	St
1	Network closet: core/distribution switches, router, patch panels, campus uplink	Decides the core design and the real bottleneck	Visit with the IT office; photograph labels	Sadat	O

#	Item to verify	Why it matters	Method	Owner	St
2	AP and ceiling hub models; cable category	Hub vs. switch changes the diagnosis; AP model shows the Wi-Fi standard	Photograph back labels and cable print	Emon	O
3	Wi-Fi and wired speed tests (busy vs. quiet times)	Evidence for the “network is slow” problem and for H1/H2	Speed tests in 201, TA office, lecture rooms, corridors	Emon	O
4	Fire devices inside each professor’s office	Decides how many detectors, alarms and sirens to add	Check all 13 offices	Islam	O
5	Fire alarm control panel – location and type	How the new system should coexist with the existing one	Site survey / facilities office	Islam	O
6	Which current Cisco devices and IoT devices our Packet Tracer version supports	Decides the stand-ins used in the simulation	Test in Packet Tracer	Hassan	O
7	AP locations on 1F and the 26 devices still in the 3D model’s “to place” tray	Completes the as-is model before the fire scenarios	Site survey, update 3D model	Pushkal	O
8	Room-number check (International Administrative Center listed as 218) and the 3F layout	Correct labels in the model and the report	Site survey	Pushkal	O

4.2 Factors of technical uncertainty

- The real bottleneck may be the campus uplink rather than W19's access layer; then new access switches alone would not fix the slowness.
- The ceiling "hub" units may be PoE switches rather than true hubs, which changes their role in the diagnosis.
- Packet Tracer may not include the newest Catalyst 9000 switches, the Catalyst 9800 WLC or Wi-Fi 6 APs, so stand-in models may be needed.
- Our fire model in the 3D tool is simplified (room-by-room spread, not smoke physics), so it shows network resilience trends, not exact fire timings.
- Code-compliant fire alarm systems are regulated; our networked alarm is designed to add notification and monitoring on top of the existing system, not to replace it.

4.3 Experiments and analysis

#	Experiment	Confirms or rejects
E1	Speed tests at busy and quiet times on 1F and 2F	H1/H2: slow wired access and thin Wi-Fi
E2	Packet Tracer baseline (as-is) vs. upgraded topology: connectivity, latency and VLAN isolation in simulation mode	Whether the new design removes the bottleneck
E3	Packet Tracer IoT fire chain: smoke detector → alert server → siren and alarm in each office	H3: networked per-office alarms work end-to-end
E4	3D fire scenarios S1–S5 (below): % of offices alerted, time to alert, single points of failure	H4: the network survives a fire
E5	Wi-Fi coverage check in the 3D model: proposed AP positions vs. room positions	Every office and corridor is inside an AP cell

Planned fire scenarios for E4:

- **S1** – fire starts in a professor's office (e.g. 226) late in the evening, professor alone.
- **S2** – fire starts in the network closet (IDF) serving 2F.

- **S3** – fire in the north/east corridor ceiling near Hub B and AP 4280.
- **S4** – power cut to one switch (tests PoE and UPS on the fire VLAN).
- **S5** – fire in 201 (AI Research Institute, 24 workstations).

5. Project Constraints

Type	Constraint	Impact on our design
Technical	Packet Tracer's device catalogue may not include the newest Cisco models	Use the closest stand-in (e.g. 3650 multilayer switch for a Catalyst 9300/9500) and list the real model in the bill of materials
Technical	Packet Tracer cannot simulate real fires or physical damage	Use the 3D model of W19 for fire scenarios; Packet Tracer for protocols and configuration
Time	Proposal @November 11, 2026, progress @November 25, 2026, completion @December 9, 2026 (about 4 weeks of build time)	Prioritise P1 goals; design freeze on @November 30, 2026
Environment	May not be allowed into restricted zones (network closet, TA office)	Ask the IT office early; if refused, design from standard assumptions and state them clearly
Environment	No real equipment to install or test	All results are simulated; costs come from public list prices
Tool	Packet Tracer runs on our laptops; the 3D model runs in a browser	Keep the 3D model lightweight; share files through Notion and GitHub
Scope	W19 1F and 2F (3F only if time allows); fire devices in the 13 professors' offices; simplified fire model	Clear boundary: we do not redesign the campus network or replace the code-compliant fire alarm panel

6. Project Goals

6.1 Technical goals

- An upgraded W19 network in Packet Tracer using a hierarchical (collapsed core + access) design with VLANs, inter-VLAN routing, DHCP and centrally managed Wi-Fi.
- A smoke detector, alarm and siren in each of the 13 professors' offices, on a dedicated fire-safety VLAN with QoS priority and an alert server.
- A fire-scenario mode in the 3D model of W19 that tests the chosen network and reports alert coverage and single points of failure.
- An itemised bill of materials and total upgrade cost.

6.2 Learning goals

- Configure Cisco switches, routers, WLC and APs ourselves (VLANs, trunks, SVIs, DHCP, ACLs, QoS).
- Apply the Cisco Campus LAN & WLAN design guide to a real building.
- Integrate IoT safety devices into an enterprise network and design for failure, not just for normal operation.
- Survey, document and divide work as a team, with the full history in Notion.

6.3 Proposed design direction

W19 has a few hundred wired and wireless devices and will need fewer than about 25–50 APs, which fits the Cisco design guide's small-to-medium campus designs. We will use a collapsed core (distribution and core in one layer) with access switches on each floor.

Layer	Real-world Cisco product (for the cost estimate)	Packet Tracer stand-in (to confirm, item 6)
Edge / WAN	Catalyst 8300 Series edge platform	ISR 4331 router
Collapsed core / distribution	Catalyst 9500 (or a stacked pair of Catalyst 9300)	3650-24PS multilayer switch (×2 for redundancy)
Access	Catalyst 9300 / 9200 PoE+ access switches with multigigabit uplinks	2960 access switches

Layer	Real-world Cisco product (for the cost estimate)	Packet Tracer stand-in (to confirm, item 6)
Wireless	Catalyst 9800-L wireless controller + Catalyst 9100-series Wi-Fi 6 APs	WLC + lightweight APs
Fire-safety IoT	IP smoke/heat detectors, alarms and sirens per office; IoT gateway	Packet Tracer IoT smoke detector, siren, alarm, MCU + IoT registration server
Servers	Alert & monitoring server, DHCP/DNS	Server-PT (IoT service, DHCP, DNS)

Draft VLAN and IP plan (10.19.0.0/16, "19" for W19):

VLAN	Purpose	Subnet
10	Faculty & staff (professors' offices, Dean, TA office)	10.19.10.0/24
20	Labs & students (201, 215–217, 1F lecture/project rooms)	10.19.20.0/24
30	Wireless clients	10.19.32.0/22
40	Fire-safety IoT (detectors, alarms, sirens) – highest QoS priority	10.19.40.0/24
50	Security (CCTV, motion sensors)	10.19.50.0/24
99	Network management	10.19.99.0/24

Fire alert chain: detector in an office → PoE access switch (VLAN 40, prioritised) → core → alert server → siren and alarm in that office and the corridor, plus a notification to every professor's PC and the TA office. The fire VLAN gets UPS-backed PoE and redundant uplinks, and an ACL allows IoT devices to talk only to the alert server.

6.4 Priorities & success criteria

Priority	Goal	Success criterion (measurable)
P1	Upgraded W19 topology working in Packet Tracer	Every end device gets an address by DHCP and reaches the servers; inter-VLAN routing works; VLAN 40 is isolated by ACL

Priority	Goal	Success criterion (measurable)
P1	Fire alarm in every professor’s office	In simulation, a triggered detector sounds the siren and alarm in its office and notifies all 13 offices and the TA office
P1	Network survives a fire (3D model)	In scenarios S1–S5, 100% of offices outside the fire zone still receive the alarm; no single failure outside the fire zone cuts any office off
P2	Full Wi-Fi coverage	Every office, lab and corridor on 1F/2F lies inside a proposed AP cell in the 3D model
P2	Cost estimate	Itemised bill of materials (quantities from the 3D model JSON) with a total
P3	Extra features	Live dashboard of fire-device status; 3F included

7. Project Schedule

7.1 Step-by-step action plan

Due	Course milestone	Our tasks	Owner
@October 28, 2026	Motivation	Team formed, ground rules signed, Packet Tracer course	All
@November 4, 2026 4 November 2026	Investigate W19 N/W	Site survey of 1F/2F (56+ photos), 2D maps, 3D model of W19	Islam, Pushkal, All
@November 11, 2026 11 November 2026	**Proposal presentation**	Finish proposal → slides (submit by @November 9, 2026) → rehearse	Pushkal, Islam, Sadot, Hassan, All
@November 18, 2026 18 November 2026	Plan & design N/W	Closet visit and speed tests (E1); confirm Packet Tracer catalogue; as-is topology; VLAN/IP plan; proposed devices placed in the 3D model	All

Due	Course milestone	Our tasks	Owner
@November 25, 2026	**Progress presentation**	Upgraded topology working (VLANs, routing, WLC, DHCP); IoT fire chain in one office (E3); 3D fire-scenario mode v1 with first results	All
@December 2, 2026	Upgrade N/W	Fire devices in all 13 offices; redundancy; run S1–S5 (E4) and fix the design; Wi-Fi coverage check (E5); cost estimate	All
@December 9, 2026	**Completion presentation**	Final .pkt file, 3D fire-scenario demo, cost estimate, Notion history	All

Weekly team meeting on Saturdays 10pm: Zoom link

7.2 Interim checkpoints

- Weekly report in the Weekly Reports database every Wednesday, in the professor's format.
- Internal review every Monday before a presentation (@November 9, 2026, @November 23, 2026, @December 7, 2026).
- Design freeze on @November 30, 2026: after this, only fixes found by the fire scenarios.
- Cross-check after every scenario run: any change made in the 3D model is mirrored in the Packet Tracer file, and vice versa.

7.3 Risks & response plan

Risk	Likelihood	Impact	Response
No access to the network closet / IT details	Medium	High	Ask the IT office this week; if refused, use the design guide's standard assumptions and state them clearly

Risk	Likelihood	Impact	Response
Needed device missing in Packet Tracer	High	Medium	Use the closest stand-in; list the real model in the BOM
3D fire-scenario mode takes longer than planned	Medium	Medium	Build a minimum version first (room-by-room spread + path check); extras only after P1 is done
Packet Tracer and 3D model drift apart	Medium	Medium	One shared device list (3D JSON export) as the source of truth; cross-check after each run
Team member unavailable	Medium	Medium	Every part has a backup owner; work is logged in Notion so others can continue
Speed tests do not show slowness	Low	Medium	Re-test at peak times; reframe the upgrade around capacity, resilience and the fire system

8. References

Official documents

- Cisco, *Campus LAN and Wireless LAN Solution Design Guide*, 2020
- Cisco, *Catalyst 8300 and 8200 Series Edge Platforms Software Configuration Guide* (IOS XE 17)
- Cisco, Catalyst 2960-X Series support page
- Cisco, *Packet Tracer Download and Installation Instructions*
- NFPA 72, *National Fire Alarm and Signaling Code* (background on fire alarm requirements)

Teaching materials & reference manuals

- E. Y. Kim, *Computer Networks PBL Class – Week 1 Introduction*, LMS, 2026

Computer Networks PBL Class - Week1(Introduction) - LMS.pptx

Computer Networks PBL Class - PBL Week2(2026 Fall) (1).pptx

- Cisco Networking Academy, Getting Started with Cisco Packet Tracer
- Packet Tracer lectures: English · Korean
- three.js documentation, threejs.org (3D model)

Papers & technical data

- I. Shemsi, “Boosting Campus Network Design Using Cisco Packet Tracer,” *International Journal of Innovative Science and Research Technology*, vol. 2, no. 11, Nov. 2017

Team data

- Team Flame Guard, W19 site survey photos and 2D survey maps (1F, 2F), Nov. 2026
- Team Flame Guard, W19 3D model and layout data, https://github.com/Pushkaltoocool/FlameGuard/